Smart Traffic Control System Using Vehicle Density Analysis

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***Abstract*—Fixed-time traffic signals give every approach the same green time regardless of demand, which wastes green on lightly used approaches and lengthens queues on busy ones. This paper presents a density-aware signal controller for an isolated four-arm junction and evaluates it in a seeded microscopic simulation against the fixed plan and four re-implemented alternatives: a Webster-optimized fixed plan, vehicle-actuated control, max-pressure control and three-level density control. Each approach receives a score that weights vehicles by class and counts a queued vehicle three times as much as a moving one; green is extended between 8 s and 45 s while the served direction's score dominates, yellow and all-red intervals are never shortened, and an emergency preemption layer keeps full change intervals. Below capacity the controller reduced mean stopped delay by 35.3–36.1% relative to the 20 s fixed plan ($p < 0.001$), but the Webster plan, actuated and max-pressure control did as well (37.5–39.7%, 37.2–39.9% and 37.3–39.8%): most of the gain comes from replacing an untuned plan. Near and above capacity, density-only switching ended greens too early and lost 18.9% of capacity; a discharge-hold rule that keeps green while the served queue is still at the stop line restored it (+9.8% relative to fixed timing) and kept the low-demand benefit. An ablation shows that the queued/moving distinction, not the class weights, drives the score's benefit. With a driver reaction time calibrated to field saturation headways the comparisons hold, provided the hold's stop-line zone is re-tuned to the wider vehicle spacing (20 m: +6.1% capacity). A sensing study shows that the present detector would make every controller worse than fixed timing near capacity. A per-step audit of 3,524 runs found no conflicting greens or shortened change intervals. Control uses simulated vehicle states: on synthetic frames the Haar-cascade detector reached only 4% precision and 8% recall.**

***Index Terms*—adaptive traffic signal control, vehicle density estimation, actuated control, max-pressure control, emergency vehicle preemption, oversaturation, microscopic traffic simulation**

I. Introduction

Signalized intersections account for a large share of the delay in urban road networks. Many junctions, especially in smaller cities, still run fixed-time plans in which each approach receives a preset green interval regardless of how many vehicles are waiting. Such plans are dimensioned for average flows; when demand fluctuates or is unevenly distributed across approaches, green time is wasted on nearly empty approaches while queues build on busy ones.

Traffic-responsive systems such as SCATS and SCOOT adjust timings from detector data, but they rely on dedicated detector infrastructure and centrally managed installations [3], [4]. Low-cost alternatives estimate density locally with infrared sensors, image-based density classes, deep detectors on camera feeds or existing surveillance cameras, and adapt green time accordingly [1], [5], [6], [8]–[10].

Motivation. These low-cost systems are usually demonstrated on a prototype or a few scenarios and report an average improvement over a fixed plan. Three questions are rarely answered: whether timing logic that varies green keeps safe change intervals, how emergency priority interacts with density-based timing, and how the controller behaves as demand approaches and exceeds capacity. A fourth is how much of a reported gain is due to the density logic at all, rather than to replacing an untuned fixed plan with any demand-responsive one.

Novelty and difference from existing approaches. Rather than proposing a new detector, this paper studies the timing logic of a density-based controller under controlled, statistically paired conditions, against re-implementations of the strategies used by existing approaches (a Webster-optimized fixed plan [2], vehicle-actuated control [23], max-pressure control [11] and the three-level density control of [1]) on identical traffic. The evaluation exposes a failure mode of density-only switching near capacity, which we correct with a discharge-hold rule taken from actuated control. We state the role of AI and machine learning plainly: the signal controller is a transparent rule, not a learned policy; machine learning enters through the pre-trained Haar-cascade detector of the vision pipeline, whose accuracy we measure, and through a data-driven calibration of the controller's parameters on held-out simulation seeds. The paper's contributions are:

- a class-weighted, queue-aware density score and a bounded green-extension controller that never shortens yellow or all-red intervals and guarantees a maximum red time for every approach;
- a discharge-hold extension that removes the capacity loss of density-only switching near saturation;

- an emergency preemption layer that wraps any phase-based controller and keeps full change intervals on entry and exit, with a per-step signal audit that checks these properties in every run;
- a reproducible evaluation: paired seeded runs at three demand levels below capacity and a sweep from 0.6 to 1.5 times the measured capacity, a comparison with four re-implemented strategies, an ablation of the density score and the hold rule, and the cost of preemption to other traffic;
- a lightweight Haar-cascade detection and tracking pipeline, evaluated against exact ground truth with detector and tracker ablations, failure cases and tracked-speed error;
- a validation of the simulated queue discharge against Highway Capacity Manual values, a reaction-time calibration that matches them, and a repeat of the main comparisons under the calibrated kinematics;
- a study of how the controllers degrade with imperfect sensing, down to the measured detector, and a data-driven calibration of the density score evaluated on held-out seeds.

All results were obtained in simulation; no hardware prototype or recorded field footage was used. The simulator, controllers and every experiment script are public, and each table can be regenerated from them (github.com/shivam-G05/Smart-Traffic-Management). Section II reviews related work, Section III describes the architecture and workflow, Section IV the models and algorithms, and Section V the experiments. Section VI discusses implications and limitations and Section VII concludes.

II. Related Work

A. Adaptive Signal Control

Fixed-time plans are commonly derived from Webster's delay-minimizing cycle formula [2]. SCATS and SCOOT adapt splits, cycle lengths and offsets across networks from loop-detector data [3], [4]. Vehicle-actuated control extends a green while detectors near the stop line keep registering vehicles and ends it when a gap appears ("gap-out"), within minimum and maximum green limits [23]. Max-pressure control chooses phases from queue differences and has provable throughput guarantees [11], and reinforcement-learning controllers have been studied widely, although they must be trained in simulation and their behavior is harder to bound [12]. The controller studied here is deliberately simpler: a local, rule-based green extension with hard limits whose behavior can be stated and checked exactly, compared directly against the Webster, actuated and max-pressure strategies in Section V.

B. Low-Cost Density-Based Systems

Nsofwa and Tembo classify image-based density into low, medium and high levels and adjust signal duration, showing in simulation that this outperforms fixed timing [1]. Kamble et al. apply YOLO-based recognition to IP-camera feeds to prioritize the busiest lane [6]; such detectors [7] are accurate but demanding on low-power hardware. Pardeshi et al. use passive infrared sensors on an ESP32 [5], and Patil et al. and Thomas et al. also adapt timing to measured density [9], [10]. Hu et al. estimate density from surveillance cameras [8], and related systems add incident detection and alerting [13], AI-based light design [14] or machine-learning traffic prediction on the Edge Impulse platform [15]. These works mainly report an improvement over fixed timing on a prototype or a few scenarios. This paper differs in three ways: it compares against strong baselines on identical traffic, it tests demand up to 1.5 times capacity, and it verifies change-interval safety at every simulation step.

C. Detection, Tracking and Emergency Priority

Viola and Jones introduced boosted cascades of Haar-like features [16], which remain attractive for embedded use because inference is cheap and needs no GPU. ByteTrack associates both high- and low-confidence detections with existing tracks [17]; trackers of this family combine a Kalman filter for motion prediction [18] with Hungarian assignment [19]. Nellore and Hancke give emergency vehicles signal priority based on visual sensing [20].

III. System Architecture and Workflow

Fig. 1 shows the software architecture. It consists of Python packages with one-directional dependencies: the simulation core imports no decision, analytics or web code, and every signal controller implements one interface (an update per time step and a phase name). The fixed-time, adaptive and comparison controllers can therefore be exchanged without changes elsewhere. The code uses Python 3.12 (NumPy, SciPy, OpenCV, FastAPI, SQLAlchemy with SQLite) and a React dashboard, and has 119 automated tests.

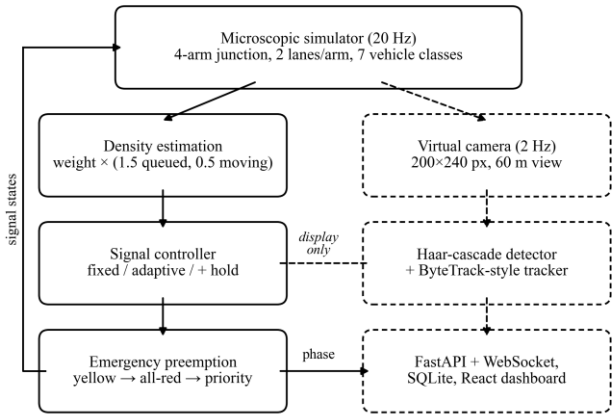


Fig. 1. Software architecture. Solid boxes form the control loop; dashed boxes form the vision pipeline, whose output is displayed and evaluated but not used for control in this study.

Simulation core: The simulator advances the junction in 50 ms steps. In each step the active controller first sets the light of every approach, and then all vehicles move.

Density and control: The density estimator scores each approach from the current vehicle states (Section IV-B). The controller uses the scores to decide when to end a green (Sections IV-C and IV-D), and the preemption wrapper overrides it when an emergency vehicle approaches (Section IV-F).

Vision pipeline: At 2 Hz a virtual camera per approach renders the 60 m before the stop line; the frame is passed to the

detector and tracker (Section IV-G). The results are shown on the dashboard and compared with ground truth, but control decisions use the simulator's vehicle states, which corresponds to ideal detection.

Backend and dashboard: A FastAPI server streams the junction state over WebSockets at 10 Hz, samples queue, delay, throughput and congestion level at 1 Hz, stores a snapshot in SQLite every 10 s, and exports reports as CSV or JSON. A React dashboard draws the junction to scale, raises a high-traffic alert when an approach stays at high density for 3 s, explains each displayed quantity, and shows a live comparison in which a second simulation with the same random seed runs another controller.

Method workflow: Fig. 2 summarizes the method. In every step the controller senses vehicle positions, speeds and classes, scores each approach, and decides whether the current green may end (Fig. 2b); the preemption layer then protects every transition with full yellow and all-red intervals before the signal heads change.

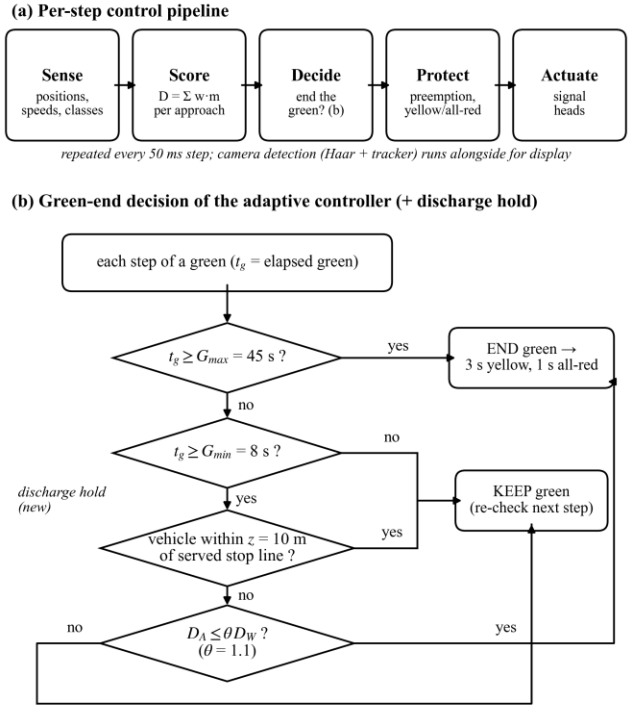


Fig. 2. Method workflow. (a) The per-step control pipeline. (b) The green-end decision; the test marked "discharge hold (new)" (Section IV-D) is the extension added in this revision, and the original controller skips it.

IV. Methodology

A. Junction and Traffic Model

The junction has four approaches, each with two lanes on a 330 m link; the stop line is at 200 m and the far edge of the junction box at 218 m. Vehicles travel straight through; turning movements, lane changes and pedestrians are not modelled. Vehicles arrive on each approach at a mean rate λ with deterministic headways (one vehicle every 1/λ s); to create shifting imbalance, every 15 s the rate of each approach is redrawn as $\lambda \cdot U(0.4, 1.6)$, with a floor of 0.02 veh/s. An arriving vehicle enters a lane at rest and is dropped if the rear of the previous vehicle is still within 8 m. For the above-capacity experiments (Section V-D) two options are enabled instead: a blocked arrival waits in an entry queue and enters as soon as there is room, so that unserved demand and its waiting time are measured, and a vehicle enters at the highest speed from which it can still stop behind its leader, so that the link entry does not cap the flow below the signal's capacity. Table I lists the vehicle classes.

TABLE I
VEHICLE CLASSES AND PARAMETERS

Class	Share (%)	Length (m)	v_{max} (m/s)	a (m/s²)	b (m/s²)	Weight w
Car	55	4.5	14	2.5	4.5	1.0
Motorcycle	25	2.0	15	3.5	5.0	0.5
Auto-rickshaw	10	3.2	11	2.0	4.0	0.8
Bus	5	10.0	10	1.2	3.0	3.0
Truck	4	9.0	9	1.0	2.8	2.5
Ambulance	0.5	5.5	16	3.0	5.0	6.0
Fire truck	0.5	10.5	12	1.5	3.5	6.0

a: maximum acceleration; b: maximum deceleration; w: density weight in (2).

Longitudinal motion follows a Gipps-type safe-speed rule [21]. A follower's speed is limited to

$$u_{safe} = [2b(g + u_l^2/(2b_l) - s_0)]^{1/2} \quad (1)$$

where g is the gap to the leader's rear, u_l and b_l are the leader's speed and maximum deceleration, b is the follower's maximum deceleration and s_0 = 2 m is the standstill gap. A position constraint prevents any overlap within a step. On yellow or red, a vehicle upstream of the stop line brakes once its braking distance reaches the distance to the line; a vehicle too close to stop continues. Rule (1) has no driver reaction time; Section V-H shows that this makes queues discharge too fast, adds a reaction time τ, and repeats the main comparisons with it.

B. Density Estimation

For approach r with vehicle set V_r, the density score is

$$D_r = \Sigma_{v \in Vr}\, w_{c(v)} \cdot m(v) \quad (2)$$

where $w_{c(v)}$ is the class weight (Table I) and $m(v) = 1.5$ if vehicle v is queued and 0.5 otherwise. A vehicle is queued when it is upstream of the stop line and slower than 0.5 m/s. The weights reflect the road space and discharge time a vehicle class needs, and emergency vehicles receive a high weight so that their approach is favored even without preemption. A queued vehicle counts three times as much as a moving one because it needs green now, whereas a moving vehicle may still be far upstream. The weights are heuristic; Section V-F measures the contribution of each part of the score. For display, an approach is labelled low ($D_r \leq 3$), medium ($3 < D_r \leq 6$) or high ($D_r > 6$).

C. Adaptive Green Extension

The controller uses a two-phase plan: north–south green, yellow (Y = 3 s), all-red (A = 1 s), then east–west green, yellow and all-red. Only the green duration adapts. Let t_g be the elapsed green time, and D_A and D_W the summed scores of the active and the waiting direction. At every step the green ends when

$$t_g \geq G_{max} \;\text{ or }\; (t_g \geq G_{min} \;\text{ and }\; D_A \leq \theta D_W) \quad (3)$$

with G_{min} = 8 s, G_{max} = 45 s and θ = 1.1. The green is thus extended while the served direction's score exceeds the waiting direction's by more than 10%. Because yellow and all-red are fixed and green is capped, the red time of any approach in normal operation is bounded by

$$R_{max} = A + G_{max} + Y + A = 50 \text{ s} \quad (4)$$

so no approach can be starved. The fixed-time baseline uses the same sequence with a 20 s green (48 s cycle).

D. Discharge Hold

Rule (3) has a weakness that the above-capacity experiments expose (Section V-D). When both directions are saturated, the vehicles moving off on green score 0.5 each while the queue growing on red scores 1.5 each, so $D_A \le \theta D_W$ holds soon after G_{min} and greens stay short; each phase change costs $Y + A = 4$ s in which no vehicle is served, so short greens lose capacity exactly when it is scarcest. The discharge hold adds the gap-out test of actuated control [23]: past the minimum, a green may only end once the served direction has gapped out, i.e. no served vehicle is within z of its stop line,

$$t_g \ge G_{max} \text{ or } (t_g \ge G_{min} \text{ and } n_z = 0 \text{ and } D_A \le \theta D_W) \quad (5)$$

where n_z is the number of served vehicles within z = 10 m upstream of the stop line. A saturated queue moving off always keeps a vehicle at the line and so keeps its green up to G_{max}, while free-flowing arrivals cross 10 m in under a second, so a cleared queue still hands over early. The zone length was chosen from 10, 15, 20 and 30 m on a separate tuning seed that is not used in any reported result. Yellow, all-red and the green limits are unchanged, so bound (4) still holds.

E. Comparison Controllers

To compare against existing approaches (Section II), four strategies were re-implemented in the same simulator. They are not the original systems: like the proposed controller they read exact vehicle states, and all share its phase sequence, its 3 s yellow and 1 s all-red, and its 8–45 s green limits, so every one obeys bound (4) and the comparison isolates the switching decision.

Webster fixed: a fixed plan whose green follows Webster's formula [2] for the mean demand of each run: $C_0 = (1.5L + 5)/(1 - Y)$, with lost time L = 8 s, Y the sum of the critical flow ratios q/s, the saturation flow s = 6000 veh/h per approach measured in the simulator, and each phase receiving $(C_0 - L)/2$, clamped to 8–45 s. It knows the true mean demand of the run, which favors it.

Actuated: past the minimum, the green ends as soon as the served direction gaps out (no vehicle within 10 m of its stop line) and a vehicle waits on the other direction [23]; no density score is used.

Max-pressure: the pressure of a phase is the number of vehicles upstream of its stop lines minus those downstream, which is zero with free exits [11]; past the minimum the controller switches as soon as the waiting phase's pressure exceeds the served one's. This also represents busiest-lane-first control [6].

Density levels: following [1], when a green starts the served direction's density is classed low, medium or high (thresholds of Section IV-B) and the green is set to 10, 20 or 30 s, without revision during the green.

F. Emergency Vehicle Preemption

Preemption is a wrapper around any phase-based controller (Fig. 3). Without an emergency vehicle it passes control to the inner controller. When an ambulance or fire truck is upstream of the junction exit, it acts as follows.

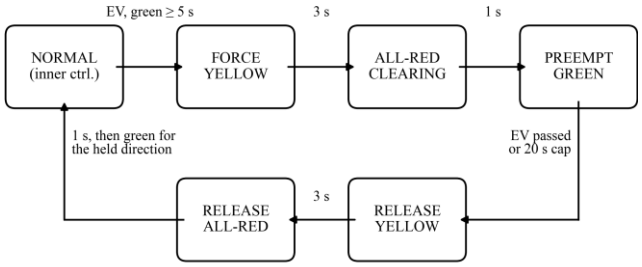


Fig. 3. Emergency preemption states. Every path from one direction's green to the other's passes through a full yellow and all-red interval.

- Entry: a conflicting green first shows a full 3 s yellow, and a yellow already showing keeps its remaining time; all approaches are then red for 1 s, or for the remainder of an all-red already under way. If the emergency vehicle's approach is already green, that green is held.
- Priority green: the emergency vehicle's direction (both opposing approaches) is green until no emergency vehicle remains before the junction exit, or for at most 20 s.
- Release: the priority green ends with a full yellow and all-red, after which the inner controller restarts at the green of the direction that was held.
- Minimum green: a green shown for less than 5 s is never cut, and an approach that already has green is served first.

Emergency vehicles are detected from the simulator state, i.e., perfect detection is assumed; in practice the input could come from vehicle-to-infrastructure messages or visual sensing. When no other preemption is active, an emergency vehicle waits at most 5 + 3 + 1 = 9 s for its priority green.

G. Vision Pipeline

For each approach a virtual camera renders a 200 × 240 pixel frame of the 60 m before the stop line (4 px/m). Vehicles appear as shaded rectangles with a windshield band on a noisy road texture with lane markings. The detector converts the frame to grayscale, equalizes its histogram, upscales it three times, and applies OpenCV's cascade classifier with a pre-trained vehicle cascade (scale factor 1.05, minimum neighbors 2, minimum size 20 px); the final-stage weight, clipped to [0, 1], serves as a relative confidence. A single cascade localizes vehicles but does not classify them, so on its own it cannot supply the class weights of (2).

The tracker keeps a constant-velocity Kalman filter on each box center and matches predicted boxes to detections with the Hungarian algorithm on an IoU cost (minimum IoU 0.15). As in ByteTrack, detections with confidence ≥ 0.15 are matched first and the remaining low-confidence detections are then matched to still-unmatched tracks; only confident detections

start tracks. A track is confirmed after two matches and deleted after three frames without one.

Data and annotation protocol: No real images are used. Frames are rendered from the simulator state, and the ground-truth box of every vehicle in view is computed from the same state, so annotation is automatic, exact and involves no manual labelling. A detection is correct when its center lies inside a ground-truth box enlarged by 4 px; each detection matches at most one vehicle, the nearest. Nothing is trained: the cascade is pre-trained, and the detector and tracker parameters were fixed during development in unseeded runs, so there is no training split; seeds 1–4 are used only for the evaluation in Section V-G. A track's speed is its Kalman velocity converted to m/s (4 px/m) and is compared with the vehicle's true speed.

V. Experimental Evaluation

A. Setup

Three demand levels below capacity were simulated: λ = 0.08, 0.12 and 0.18 veh/s per approach (1152, 1728 and 2592 veh/h offered to the junction). For each level and controller, ten random seeds were run for 1800 s with a 50 ms step. A seed gives every controller the same arrival sequence, so runs form matched pairs. The first 60 s are a warm-up, and delay statistics use the vehicles entering between 60 s and 1650 s, all of which left the network before the end of their run. Stopped delay is the time a vehicle spends upstream of the stop line below 0.5 m/s; because vehicles enter at rest it includes a start-up component of 0.15–0.55 s that is the same for all controllers. Per-seed values are compared with paired t-tests over the ten seeds, and 95% confidence intervals are given. Preemption is enabled for all controllers unless stated.

For the above-capacity experiments, the capacity of each controller was first measured by flooding every approach (1.5 arrivals/s; five seeds); demand was then set to 0.6, 0.8, 0.9, 1.0, 1.2 and 1.5 times the measured fixed-time capacity, with ten paired seeds per level. In total 3,524 runs (1,797 simulated hours) were made. Separate seeds were used for tuning (seed 1000), capacity measurement (101–105) and the reported results (1–10).

B. Signal Safety Audit

Every step of every run was checked for conflicting greens, green-to-red transitions without yellow, yellows shorter than 3 s and all-red intervals shorter than 1 s before a conflicting green. Across all 3,524 runs, covering all seven controllers and 1,830 preemption events in the main runs alone, none of these violations occurred. An earlier version of the preemption layer returned control to the inner controller without a release interval; this audit exposed the resulting skipped yellows, which led to the release states of Fig. 3.

C. Comparison Below Capacity

Table II compares all controllers at the three demand levels. The proposed adaptive controller reduced mean stopped delay by 35.3–36.1% relative to the 20 s fixed plan, with lower delay in all ten seeds at every level ($p < 0.001$), and the 95th-percentile delay fell by 37.1–41.2%. However, the Webster plan, actuated control and max-pressure control achieved the same or slightly lower delay (37.5–39.7%, 37.2–39.9% and 37.3–39.8% below the fixed plan). Relative to the adaptive controller their delay differed by between −6.8% and −2.9%. Below capacity, therefore, most of the gain comes from replacing an untuned 20 s plan with shorter, demand-responsive greens (mean adaptive green 9.5 s), which any of these strategies provides. Three-level density control was worse than the fixed plan at medium and high demand (+29.0% and +28.9%), because its greens of 20–30 s for medium and high density outlast the queue. The discharge hold left delay essentially unchanged at these levels (between −1.2% and +2.5% relative to the adaptive controller). Throughput changed by at most 0.9% because all three levels are far below capacity.

TABLE II
Controllers below capacity (preemption enabled)

Controller	Delay low (s)	Delay medium (s)	Delay high (s)	High vs fixed	95th pct. high (s)	Green (s)
Fixed (20 s)	6.96 ± 0.17	7.20 ± 0.20	7.36 ± 0.22	—	23.6 ± 0.3	19.3
Webster fixed	4.29 ± 0.18	4.34 ± 0.15	4.60 ± 0.29	−37.5%	13.9 ± 1.5	8.7
Actuated	4.19 ± 0.21	4.36 ± 0.13	4.63 ± 0.17	−37.2%	13.5 ± 0.4	8.8
Max-pressure	4.22 ± 0.14	4.33 ± 0.17	4.62 ± 0.18	−37.3%	13.9 ± 0.6	8.9
Density levels	7.35 ± 0.37	9.29 ± 0.33	9.49 ± 0.15	+28.9%	32.0 ± 0.3	26.9
Adaptive	4.49 ± 0.10	4.60 ± 0.14	4.76 ± 0.12	−35.3%	14.8 ± 1.0	9.5
Adaptive + hold	4.44 ± 0.10	4.62 ± 0.16	4.89 ± 0.23	−33.7%	15.3 ± 1.5	9.7

Mean ± 95% confidence interval over ten paired seeds. All changes relative to the fixed plan are significant ($p < 0.001$) except density levels at low demand ($p = 0.06$). Green: mean green at high demand.

D. Operation Near and Above Capacity

The three demand levels of Table II are only 12–26% of the junction's measured fixed-time capacity, so they cannot show congestion. Table III gives the capacity of each controller; these runs and Table IV use preemption off, because emergency vehicles make up 1% of arrivals and at these demands trigger a preemption every 15–30 s, which would otherwise dominate the timing of every controller.

TABLE III
Capacity of each controller

Controller	Capacity, preempt. off (veh/h)	vs fixed	Mean green (s)	Capacity, preempt. on (veh/h)
Fixed (20 s)	10008 ± 89	—	20.0	9701 ± 197
Webster fixed	11168 ± 32	+11.6%	45.0	9977 ± 224
Actuated	10971 ± 75	+9.6%	37.3	10018 ± 223
Max-pressure	8128 ± 136	−18.8%	8.1	8876 ± 296
Density levels	10637 ± 55	+6.3%	30.0	9976 ± 193
Adaptive	8112 ± 155	−18.9%	8.1	8861 ± 321
Adaptive + hold	10988 ± 41	+9.8%	37.6	10012 ± 222

Every approach flooded at 1.5 arrivals/s; five seeds; throughput after the 60 s warm-up.

The proposed controller lost 18.9% of capacity relative to fixed timing ($p < 0.001$): with both directions saturated its greens fell to the 8 s minimum (mean 8.1 s), as explained in

Section IV-D. With the discharge hold the mean saturated green rose to 37.6 s and capacity to +9.8% relative to fixed timing. Max-pressure control suffered the same loss as the original rule (−18.8%), since moving vehicles also leave its pressure count as the queue discharges, whereas actuated control (+9.6%) and the Webster plan (+11.6%), which runs the maximum green above capacity, did not.

TABLE IV
DELAY INCLUDING VEHICLES STILL WAITING (S), PREEMPTION OFF

Controller	0.6×	0.8×	0.9×	1×	1.2×	1.5×
Fixed (20 s)	6.4	7.7	8.9	12.2	115.3	312.2
Webster fixed	4.8	8.2	11.5	17.1	50.6	240.4
Actuated	4.4	6.5	8.1	10.6	52.8	245.7
Max-pressure	4.6	9.7	25.6	132.0	292.9	438.8
Adaptive	4.6	10.1	41.5	148.7	298.2	440.1
Adaptive + hold	4.3	6.4	8.2	10.6	48.7	245.6

Demand as a multiple of the fixed-time capacity; mean over ten paired seeds. Vehicles still waiting at the end of a run count with the delay accrued so far, so values above capacity are lower bounds.

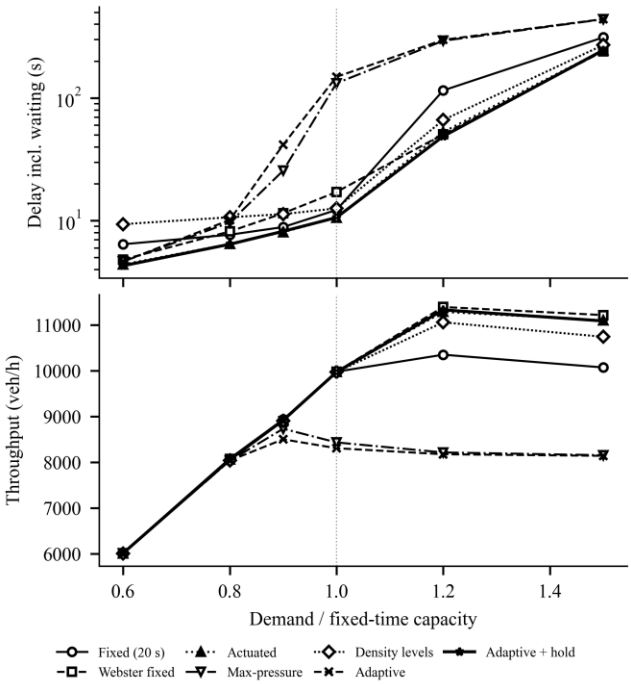


Fig. 4. Delay (top, log scale) and throughput (bottom) against demand, preemption off. The dotted line marks the fixed-time capacity.

Table IV and Fig. 4 show the consequence. The original adaptive controller was better than fixed timing only at 0.6× capacity and worse from 0.8× on: at capacity its delay was 148.7 s against 12.2 s for the fixed plan, its queue grew by 28.9 vehicles per minute, and queues reached the start of the link during 78% of approach-time. With the discharge hold, delay was below fixed timing at every level (−33.2%, −16.3%, −7.6%, −13.3%, −57.7%, −21.3% from 0.6× to 1.5×; the difference at 1.0× was not significant, $p = 0.14$) and throughput above capacity rose by +9.5% and +10.1% at 1.2× and 1.5×. Among the comparison strategies, actuated control behaved similarly to the hold controller (10.6 s at capacity, 52.8 s at 1.2×, against 10.6 s and 48.7 s), and max-pressure control degraded like the original rule. With preemption enabled the same pattern held with smaller margins (capacity −8.7% for the original rule and +3.2% with the hold), because preemption also cuts the hold's long greens.

Steady demand shows capacity but not how a controller manages a congestion episode. In a further 60-minute experiment, demand rose from 0.6 to 1.2 times the fixed-time capacity (ramping over 10–15 min), stayed there for 15 min and fell back to 0.6 by 35 min (Fig. 5; preemption off, ten paired seeds; the Webster plan timed for the average demand). The queue, including vehicles waiting to enter, peaked at 495 vehicles under fixed timing, 255 with the discharge hold, 254 with actuated control and 1214 with the original adaptive rule. After the peak, the queue returned to its pre-peak level in 13.9 min under fixed timing and 7.3 min with the hold, whereas the original rule did not recover within the hour and max-pressure control 28.7 min (9 of 10 runs did not recover). Approach entries were blocked for 68 approach-minutes under fixed timing, 36 with the hold and 172 with the original rule. During the peak the hold served 11394 veh/h, against 10574 for fixed timing and 8244 for the original rule. Relative to fixed timing, total delay changed by −57.3% with the hold, −57.6% with actuated control and +356.1% with the original rule. Managing a congestion episode therefore depends above all on not shortening greens while queues are saturated.

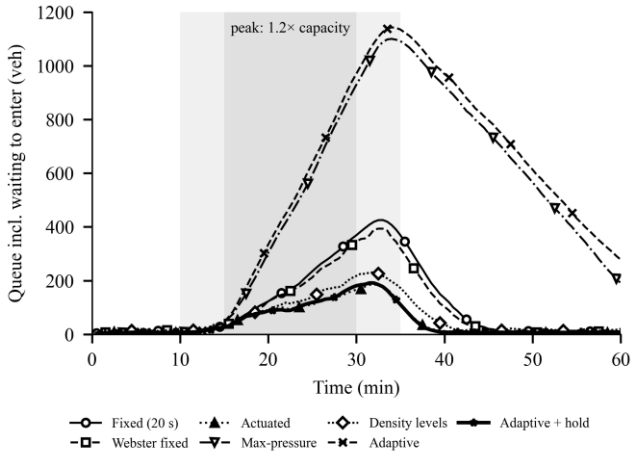


Fig. 5. Queue including vehicles waiting to enter during a 60-minute peak period (mean of ten seeds, 2-minute moving average, preemption off). Shading: demand ramping (light) and at 1.2 times the fixed-time capacity (dark).

E. Emergency Preemption

Emergency vehicles made up 1% of arrivals and triggered 12.4–23.4 preemptions per hour under adaptive control (Table V). Preemption reduced their mean stopped delay to the start-up component of the metric under every controller. The cost to other traffic depends on the controller: it is small under fixed timing, and larger under adaptive control because each preemption discards the density-responsive timing and restarts the plan at a fixed phase. Emergency vehicles are far more frequent here than in normal traffic, so the aggregate cost at a real junction would be expected to be smaller.

TABLE V
EFFECT OF EMERGENCY PREEMPTION

Demand	Controller	EVs off/on	EV delay (s) off→on	Other delay (s) off→on	Other change (p)
Low	Fixed (20 s)	58/58	7→0.28	7.06→7.04	−0.2% (0.86)
Low	Adaptive	58/58	2.42→0.28	4.34→4.54	+4.5% (< 0.001)
Low	Adaptive + hold	58/58	2.5→0.28	4.32→4.49	+3.8% (0.003)
Medium	Fixed (20 s)	75/74	8.87→0.28	7.05→7.27	+3.1% (0.08)
Medium	Adaptive	72/72	3.65→0.35	4.25→4.64	+9.3% (< 0.001)
Medium	Adaptive + hold	74/73	3.77→0.3	4.35→4.66	+7.2% (0.01)
High	Fixed (20 s)	103/111	6→0.42	7.28→7.43	+2.1% (0.17)
High	Adaptive	103/105	4.18→0.51	4.22→4.81	+13.8% (< 0.001)
High	Adaptive + hold	103/103	3.7→0.57	4.41→4.93	+11.8% (< 0.001)

EV: emergency vehicles measured, pooled over ten seeds (EV delay: Mann–Whitney U test $p \le 0.002$). Other change: paired change in the mean delay of all other vehicles, paired t-test p-value in parentheses.

F. Ablation

Table VI changes one part of the controller at a time, on the same paired seeds. Counting vehicles without weights or multipliers raised delay by 22.7% and 26.6% at medium and high demand. Removing the parts separately shows where this comes from: without the queued/moving distinction delay rose by 35.1% and 33.0%, whereas setting every class weight to 1 changed it by −3.9% and −3.2%. The benefit of the score therefore comes from distinguishing queued from moving vehicles, not from the class weights, which did not help as set (Section V-J calibrates them). Raising θ to 1.5 lowered delay slightly (−4.9% and −4.1%) and θ = 1.0 made no difference; the discharge hold, with any zone length, did not change delay below capacity, where its benefit is not needed.

TABLE VI
ABLATION OF THE ADAPTIVE CONTROLLER

Variant	Demand	Mean delay (s)	Change	p	Green (s)
Adaptive (reference)	medium	4.60 ± 0.14	—	—	9.5
Unweighted count (weights and multipliers = 1)	medium	5.65 ± 0.16	+22.7%	< 0.001	14.1
No class weights (all classes = 1)	medium	4.42 ± 0.14	−3.9%	0.07	8.9
No queued/moving distinction (multipliers = 1)	medium	6.22 ± 0.21	+35.1%	< 0.001	15.1
theta = 1.0	medium	4.66 ± 0.13	+1.3%	0.39	10.0
theta = 1.5	medium	4.38 ± 0.13	−4.9%	0.005	8.8
+ discharge hold, 10 m zone	medium	4.62 ± 0.16	+0.4%	0.82	9.7
+ discharge hold, 20 m zone	medium	4.66 ± 0.14	+1.3%	0.44	9.8
+ discharge hold, 30 m zone	medium	4.61 ± 0.12	+0.2%	0.92	10.5
Preemption off	medium	4.24 ± 0.09	−7.8%	< 0.001	9.2
Fixed 20 s (for reference)	medium	7.20 ± 0.20	+56.5%	< 0.001	19.6
Adaptive (reference)	high	4.76 ± 0.12	—	—	9.5
Unweighted count (weights and multipliers = 1)	high	6.03 ± 0.19	+26.6%	< 0.001	14.3
No class weights (all classes = 1)	high	4.61 ± 0.15	−3.2%	0.04	9.1
No queued/moving distinction (multipliers = 1)	high	6.34 ± 0.15	+33.0%	< 0.001	15.0
theta = 1.0	high	4.75 ± 0.14	−0.3%	0.84	9.8
theta = 1.5	high	4.57 ± 0.14	−4.1%	0.06	8.9
+ discharge hold, 10 m zone	high	4.89 ± 0.23	+2.5%	0.17	9.7
+ discharge hold, 20 m zone	high	4.87 ± 0.15	+2.2%	0.06	10.0
+ discharge hold, 30 m zone	high	4.94 ± 0.19	+3.6%	0.09	10.9
Preemption off	high	4.22 ± 0.06	−11.4%	< 0.001	8.9
Fixed 20 s (for reference)	high	7.36 ± 0.22	+54.5%	< 0.001	19.3

Preemption enabled unless stated; ten paired seeds per row; change relative to the reference at the same demand.

G. Vision Pipeline: Accuracy, Failure Cases and Ablation

The vision pipeline was evaluated on seeds 1–4 (9,600 frames, 14,874 vehicle instances in view) with the protocol of Section IV-G. On these synthetic frames the generic cascade performed poorly: precision was 4.1% and recall 8.2% (Table VII). The failure cases are systematic. 98.6% of the false positives recur at the same spot in at least 10% of a camera's frames, i.e. the cascade fires on fixed background texture rather than on vehicles. Recall varies by class (car 7.1%, motorcycle 11.8%, auto 7.6%, bus 7.2%, truck 7.7%, fire_truck 3.7%, ambulance 14.3%) and by distance from the stop line (0-20 m 5.4%, 20-40 m 16.9%, 40-60 m 17.4%); recall is lowest near the stop line, where queued vehicles stand close together. Without upscaling, or without histogram equalization, the cascade found essentially nothing. A 2× instead of 3× upscale raised precision to 63.3% but cut recall to 3.7%, so no setting of this generic cascade gives both.

TABLE VII
DETECTOR ACCURACY AND ABLATION (SYNTHETIC FRAMES)

Detector variant	Precision (%)	Recall (%)	Detections per frame
Default (3x upscale, equalized)	4.1	8.2	3.30
2x upscale	63.3	3.7	0.09
No upscaling	—	0.0	0.00
No histogram equalization	—	0.0	0.00

Pooled over seeds 1–4; a detection is correct when its center lies inside a ground-truth box enlarged by 4 px. —: too few detections for a meaningful precision.

Tracking was evaluated on the default detector's output (Table VIII). Only 0.6% of the vehicle instances in view carried a confirmed track, and 1.5% of the vehicles were confirmed at least once. For confirmed tracks on real vehicles, the tracked speed was compared with the true speed: the mean absolute error was 0.53 m/s over 82 estimates. This is not a good result: the vehicles concerned were moving at only 0.56 m/s on average, because tracks are confirmed almost only on nearly stationary, queued vehicles, so the error is as large as the speed itself and the pipeline provides no usable kinematics for moving traffic. Dropping the low-confidence second matching stage or the motion model changed coverage little, because the limiting factor is the detector's recall, not the association; confirming tracks after one match raised coverage but also confirmed single-frame false positives. These results are why the evaluated controllers read vehicle states from the simulator: a class-aware detector trained on the target imagery is needed before detections can drive control, and accuracy on real

footage is unknown. On an Intel Core i5-12450H laptop CPU, detection took about 137 ms per frame in a single process, whereas a control decision took 0.01 ms; a Raspberry Pi 4 is estimated, not measured, to be 5–10 times slower.

TABLE VIII
TRACKING QUALITY AND TRACKED-SPEED ERROR

Tracker variant	Coverage (%)	Vehicles confirmed (%)	ID switches /100 veh.	Speed MAE (m/s)
ByteTrack-style (two-stage, Kalman)	0.6	1.5	0.2	0.53
Single stage (low-confidence dropped)	0.5	1.0	0.2	0.16
No motion model (last box)	0.6	1.5	0.2	n/a
Confirm after 1 match	2.3	35.9	6.6	0.53

Coverage: share of vehicle instances in view with a confirmed track. Speed MAE: confirmed tracks on real vehicles with at least three matches; n/a where the variant has no motion estimate.

H. Validation of the Simulated Kinematics

To check the simulator's kinematics against external values, queue discharge was measured as at a real junction: the time between successive queued vehicles crossing the stop line after the green starts, in saturated fixed-time runs (five seeds, greens with at least eight queued vehicles in the lane). The headway from the fifth vehicle on gives the saturation headway, and the extra time of the first four the start-up lost time. The Highway Capacity Manual's base values are 1900 passenger cars per hour per lane (a 1.89 s headway) and a start-up lost time of about 2 s [24]. The model used so far discharges cars every 0.51 s (7,102 veh/h/lane; Table IX): without a reaction time a follower may trail by only the 2 m standstill gap at any speed. Adding a driver reaction time τ to the safe speed (1), as in Krauss's model [25], and setting τ = 1.6 s gives 1.92 s (1,877 veh/h/lane) for cars, within 1.2% of the reference. The calibrated model still has no extra delay when a queue starts to move, so its start-up lost time (-0.8 s) is lower than in the field.

TABLE IX
QUEUE DISCHARGE AGAINST FIELD REFERENCE VALUES

Model	Headway, cars (s)	Sat. flow, cars (veh/h/ln)	Sat. flow, all (veh/h/ln)	Start-up lost time (s)
Simulator, τ = 0 (used above)	0.51	7,102	3,164	2.1
Simulator, τ = 1.6 s (calibrated)	1.92	1,877	1,798	-0.8
HCM base values [24]	1.89	1,900	—	≈ 2.0

Saturated fixed-time runs, five seeds; saturation headway averaged from the fifth queued vehicle on.

All comparisons were then repeated with the calibrated model (τ = 1.6 s; the Webster plan recomputed with the calibrated saturation flow of 3600 veh/h per approach). Table X summarizes them. Below capacity, the reductions relative to the fixed plan were 33.7–35.4% for the adaptive controller and 31.9–39.2%, 35.0–39.5% and 34.7–38.9% for the Webster plan, actuated and max-pressure control. The calibrated capacity of the fixed plan was 6,148 veh/h; relative to it, the original adaptive rule reached −14.6% and the Webster plan +6.3%. The discharge hold, with the 10 m zone chosen for the default model, reached only −3.7% (mean saturated green 17.4 s): with a reaction time, vehicles leaving a queue spread out to gaps of about 20 m at speed, so a 10 m zone empties between them and the green gaps out while the queue is still discharging. The zone is therefore a kinematics-dependent parameter, as the passage time of a real actuated controller is set for local conditions. Re-tuned for the calibrated model on the tuning seed (20 m), the discharge hold reached +6.1% capacity relative to fixed timing and actuated control +6.1%, against +6.3% for the Webster plan (Table X).

TABLE X
MAIN COMPARISONS REPEATED WITH CALIBRATED KINEMATICS

Controller	Delay vs fixed, low	medium	high	Capacity vs fixed
Webster fixed	−38.0%	−39.2%	−31.9%	+6.3%
Actuated	−39.4%	−39.5%	−35.0%	−3.9%
Max-pressure	−38.9%	−38.7%	−34.7%	−14.5%
Density levels	+7.4%	+29.1%	+31.4%	+4.1%
Adaptive	−35.2%	−35.4%	−33.7%	−14.6%
Adaptive + hold	−35.5%	−35.0%	−33.7%	−3.7%
Actuated, zone 20 m	−39.0%	−38.5%	−32.4%	+6.1%
Adaptive + hold, zone 20 m	−34.1%	−34.4%	−32.0%	+6.1%

τ = 1.6 s; ten paired seeds per demand level (preemption on) and five capacity seeds (preemption off). Gap-out controllers use the 10 m zone unless a re-tuned zone is stated.

I. Imperfect Sensing

All results so far assume that the controller knows every vehicle's position, speed and class. To measure how much this matters, the vehicle-reading controllers were run behind a sensor model that reports only what a camera would: vehicles within 60 m of the stop line, refreshed at 2 Hz, without classes (as a single cascade), with each vehicle detected independently with a given recall, and optionally with phantom detections at fixed spots, like the cascade's static false positives. The last profile uses the measured pipeline of Section V-G (recall 8%, 3 phantoms). Emergency preemption still reads the true emergency flags; ten paired seeds per cell. Tables XI and XII give the results at the paper's medium demand and at 0.9 times capacity.

TABLE XI
DELAY UNDER IMPERFECT SENSING (S), MEDIUM DEMAND

Sensing	Actuated	Max-pressure	Adaptive	Adaptive + hold
Exact states	4.4	4.3	4.6	4.6
Camera (60 m, 2 Hz, no classes)	4.4 (+1.4%)	4.3 (+0.4%)	4.3 (−5.9%)	4.4 (−5.4%)
recall 90%	4.4 (+0.4%)	4.4 (+1.0%)	4.3 (−5.6%)	4.3 (−6.3%)
recall 70%	4.3 (−0.1%)	4.3 (+0.2%)	4.3 (−5.4%)	4.4 (−5.1%)
recall 50%	4.4 (+0.4%)	4.3 (−0.7%)	4.3 (−5.7%)	4.3 (−6.2%)
recall 30%	4.4 (+0.3%)	4.4 (+0.5%)	4.4 (−5.3%)	4.4 (−5.6%)
2 phantoms per approach	11.9 (+172.2%)	4.3 (+0.4%)	4.3 (−5.7%)	11.9 (+156.8%)
Measured Haar (recall 8%, 3 phantoms)	11.9 (+173.1%)	4.7 (+8.1%)	4.3 (−5.7%)	11.9 (+157.6%)

Change relative to the same controller with exact states in brackets. Fixed plan (needs no sensing): 7.2 ± 0.2 s.

TABLE XII
DELAY UNDER IMPERFECT SENSING (S), 0.9 TIMES CAPACITY

Sensing	Actuated	Max-pressure	Adaptive	Adaptive + hold
Exact states	8.1	27.2	35.6	7.8
Camera (60 m, 2 Hz, no classes)	8.5 (+4.6%)	30.1 (+10.6%)	45.0 (+26.5%)	8.5 (+8.4%)
recall 90%	8.4 (+3.5%)	32.3 (+18.6%)	45.4 (+27.5%)	8.3 (+5.6%)
recall 70%	9.0 (+10.9%)	36.4 (+33.8%)	45.4 (+27.5%)	9.4 (+19.3%)
recall 50%	11.3 (+39.8%)	33.0 (+21.2%)	45.4 (+27.5%)	14.4 (+83.9%)
recall 30%	20.6 (+154.0%)	35.7 (+30.9%)	48.6 (+36.5%)	23.2 (+195.8%)
2 phantoms per approach	15.1 (+86.2%)	30.1 (+10.6%)	45.0 (+26.5%)	15.1 (+92.0%)
Measured Haar (recall 8%, 3 phantoms)	163.3 (+1915.8%)	33.2 (+21.9%)	45.2 (+27.0%)	162.9 (+1973.7%)

Delay includes time waiting to enter the road; preemption off. Fixed plan: 8.8 ± 0.2 s.

Below capacity the controllers tolerated missed detections well, because missing vehicles only shortens greens, which costs little when queues are short. Near capacity, sensing mattered: the discharge-hold controller's delay rose from 7.8 s with exact states to 8.5 s with a perfect camera view and 14.4 s at 50% recall, against 8.8 s for the fixed plan. Phantom detections were the most damaging error for the gap-out rule: a phantom inside the 10 m zone never moves, so the green never gaps out and runs to the maximum (medium-demand delay 11.9 s, +64.8% relative to fixed timing). With the measured pipeline, every vehicle-reading controller was worse than the fixed plan near capacity (discharge hold 162.9 s against 8.8 s). At medium demand max-pressure and adaptive still beat the fixed plan, because phantoms appear on both directions and largely cancel in their comparisons, whereas actuated and adaptive + hold were held at long greens by phantoms in the stop-line zone (11.9 s against 7.2 s). Closing the loop with the present detector would therefore harm traffic; a detector must at least suppress static false positives near the stop line, and its recall matters most near capacity.

J. Calibration of the Density Weights

The class weights and multipliers of (2) were set by hand. To calibrate them from data, a random search over the score parameters of the discharge-hold controller (four class weights and the emergency-vehicle weight, log-uniform 0.25–4 and 1–10; the queued/moving ratio, 1–12; and θ, 1.0–1.6; the car weight stays 1 because rule (3) compares ratios) evaluated 14 candidates on tuning seeds 1001–1002 at medium demand, high demand and 0.9 times capacity, minimizing delay relative to the hand-set weights. The best candidate, the hand-set weights and a plain vehicle count were then compared on the held-out seeds 1–10 (Table XIII).

TABLE XIII
CALIBRATED SCORE PARAMETERS AND HELD-OUT DELAY

Parameter / condition	Hand-set	Calibrated	Plain count
Motorcycle weight	0.50	1.89	1.00
Auto-rickshaw weight	0.80	3.02	1.00
Bus weight	3.00	1.67	1.00
Truck weight	2.50	0.93	1.00
Emergency-vehicle weight	6.00	1.91	1.00
Queued/moving ratio	3.00	2.94	1.00
Switching ratio θ	1.10	1.37	1.10
Delay, medium (s)	4.62	4.53 (−2.0%, p 0.374)	5.69 (+23.2%)
Delay, high (s)	4.89	4.64 (−4.9%, p 0.036)	5.99 (+22.7%)
Delay, 0.9x capacity (s)	7.85	8.03 (+2.2%, p 0.014)	8.90 (+13.4%)

Delays on held-out seeds 1–10 (ten paired seeds); changes and p relative to the hand-set weights.

On the held-out seeds the calibrated score changed delay by −2.0% (medium), −4.9% (high), +2.2% (0.9x capacity) relative to the hand-set weights; the improvement was significant for high and delay rose significantly for 0.9x capacity. The plain count remained clearly worse, confirming Table VI. The calibrated parameters are fitted to this simulator's traffic and must be re-fitted against field data before deployment; the search is a simple, reproducible way to do so, not a learned controller.

VI. DISCUSSION

A. Implications

Two results qualify the usual claim that density-based control beats fixed timing. Below capacity, the gain over an untuned 20 s plan is real but not specific to density scoring: a Webster plan tuned to the demand, actuated control and max-pressure control achieve the same. Near capacity, density-only switching is worse than fixed timing, because it ends greens while the served queue is still discharging. With the discharge hold, the proposed controller matches the best strategies below capacity and comes close to the highest capacity measured (the Webster plan's, +11.6%), while keeping the properties that motivated it: its decisions follow a rule that can be stated in one line, its red time is bounded in closed form, and its safety is audited at every step. Its results are, however, almost identical to plain actuated control, which uses no density score at all: in this setting the gap-out test, not the score, does most of the work. The score's remaining role is to decide when a cleared green should hand over, where Table VI shows that the queued/moving distinction matters. Emergency priority and adaptive control interact: resuming after preemption with the higher-scoring direction, rather than always the held one, is a direct improvement to test.

B. Limitations

- Simulation only. The model has no turning movements, lane changes, pedestrians or driver variability and has not been calibrated against field data, so absolute delays are not field predictions.
- Kinematics. The model behind the main results has no driver reaction time, so its queues discharge much faster than real ones (Table IX). A reaction time of 1.6 s matches the field saturation headway, and Section V-H repeats the comparisons with it, but the calibrated model still lacks start-

up delay, so its start-up lost time is too low; with real start-up lost time, short greens would cost more, so the original rule's capacity loss is if anything understated. The gap-out zone of the discharge hold must also suit the local vehicle spacing: 10 m for the default model, 20 m for the calibrated one, as the passage time of a field actuated controller is set on site.

- Arrival model. Arrivals follow deterministic headways at a rate redrawn every 15 s, not a Poisson process, and emergency vehicles are far more frequent than in real traffic.
- Comparison controllers. The baselines are re-implementations of the published strategies with ideal sensing, not the original systems; the Webster plan is given the true mean demand of each run.
- Sensing. The main results use exact vehicle states, and emergency vehicles are detected perfectly. Section V-I shows that the controllers need a detector far better than the present Haar pipeline, in particular one without static false positives near the stop line. The vision pipeline was tested only on synthetic frames; real footage with lighting changes, occlusion and weather is harder.
- Parameters. The calibrated score parameters (Section V-J) are fitted to this simulator, not to field data; the congestion thresholds used for display were set by hand.

A planned Raspberry Pi 4 deployment, with one camera per approach and relay-driven signal heads, is estimated at roughly INR 6,000–8,000 in components; this cost and the hardware timing in Table VII are estimates that must be verified on actual hardware.

VII. Conclusion

This paper presented and evaluated a density-aware traffic signal controller with a class-weighted, queue-aware density score, bounded green extension, a discharge-hold rule and an emergency preemption layer that keeps full change intervals. Compared on identical seeded traffic with a fixed plan and four re-implemented strategies, it reduced mean stopped delay below capacity by 35.3–36.1% relative to the 20 s fixed plan, as did a Webster-optimized plan, actuated and max-pressure control. Near capacity the original density rule lost 18.9% of capacity; the discharge hold restored and exceeded the fixed plan's capacity (+9.8%) and gave lower delay than fixed timing from 0.6 to 1.5 times capacity (not significantly at 1.0×). Under calibrated kinematics these findings held once the hold's zone was re-tuned (20 m). No run violated a change interval. The queued/moving distinction, not the class weights, accounted for the benefit of the density score.

Future work will close the loop between detection and control with a class-aware detector trained on real footage, calibrate the class weights against local passenger-car-unit factors, test the controller in an established simulator such as SUMO [22] with turning traffic and stochastic arrivals, improve resumption after preemption, and measure its performance on embedded hardware at a real junction.

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